



Component Defect Detection After Board Mounting

The fact that a component is mounted on a board does not necessarily reduce its ability to be imaged and checked with ultrasound. Here is how it can be done

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As they progress from raw materials to ready-to-mount devices, electronic components encounter various mechanical and thermal stresses. These stresses are capable of initiating internal structural anomalies such as cracks, non-bonds, and delaminations.

Such anomalies threaten the long-term performance of a component and of the system in which it has been installed. Assemblers try hard to identify suspect and defective components and remove them before they are mounted onto circuit boards. The mounting process itself can also be cause for concern because the components are subjected to mechanical and thermal stresses that can induce defects in components that were defect-free.

Structural anomalies can be found without harming the component package by using high-frequency ultrasound imaging to 'look inside it.' In this process, focused pulses enter the surface of the component and echoes return from internal interfaces. As a pulse travels through a component, these events may occur:

- When the pulse strikes an interface between two bonded solids, a predictable portion of energy is reflected back to the

transducer. The other portion goes deeper into the component.

- As the pulse travels through a homogeneous material, no echoes are generated.
- If the pulse strikes an air gap, almost all of it is reflected. None crosses the interface because ultrasound at these frequencies does not travel through air.

An acoustic micro imaging tool, such as a Nordson SONOSCAN C-mode Scanning Acoustic Microscope (C-SAM), scans over the area of the part, produces the pulses, collects the echoes, and assembles the data into an image. The tool's transducer is immersed in water to couple the ultrasound to the top surface of the component. The transducer moves swiftly back and forth across the part and pulses ultrasound at up to 25,000 x-y locations per second, receiving the echo from one pulse before launching the next pulse. Frequently, components are screened in trays to identify for removal those with internal defects, but components mounted on boards—the subject here—can be imaged after reflow to look for defects.

The amplitude of an echo carries information about the interface from which it came, and this is used to assign a specific colour to its pixel. Usually, the operator of the tool does not attempt to image the entire thickness of the component, but only the depths of interest.

If all the echoes in the acoustic image correspond to known internal interfaces in x-y location and amplitude, the part is judged to have no

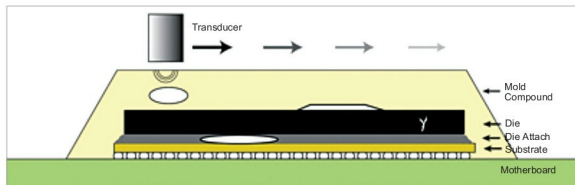


Fig. 1: Side view diagram of a BGA being imaged by ultrasound